

BANGLADESH RESEARCH AND EDUCATION NETWORK

NOC MONTHLY REPORT

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Prepared by
BdREN NOC



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1. COORDINATION ISSUES

Processed 565 mail updates between BdREN NOC and member institution.

2. LINK AVAILABILITY

Table 1: Institution Wise Link Availability

Name of Institutes	Outage (Minute)	Maintenance (Minute)	Availability (w/o Maintenance)	Availability (with Maintenance)
AUB	468		100.00%	100.00%
AUST			98.95%	98.95%
BARI			100.00%	100.00%
BAU			100.00%	100.00%
BDU (Dhaka)	52		100.00%	100.00%
BDU (Gazipur)			100.00%	100.00%
BFRI			100.00%	100.00%
BLRI			100.00%	100.00%
BoU			99.87%	99.87%
BRAC			100.00%	100.00%
BRRI			100.00%	100.00%
BRUR			100.00%	100.00%
BSMMU			100.00%	100.00%
BSMRAU			100.00%	100.00%
BSMRSTU			100.00%	100.00%
BU			100.00%	100.00%
BUBT			100.00%	100.00%
BUET			100.00%	100.00%
BUP			100.00%	100.00%
BUTex			100.00%	100.00%
CoU	1023		100.00%	100.00%
CU			100.00%	100.00%
CUET			100.00%	100.00%
CVASU			100.00%	100.00%
DIU			100.00%	100.00%
DU			100.00%	100.00%
DUET			100.00%	100.00%
EU			100.00%	100.00%
HSTU			100.00%	100.00%
ICMAB			100.00%	100.00%
IU			100.00%	100.00%
IUB			100.00%	100.00%
IUBAT			100.00%	100.00%
IUT			100.00%	100.00%
JKKNIU			100.00%	100.00%
JnU			97.71%	97.71%
JU			100.00%	100.00%
JUST			100.00%	100.00%

KU	1989		95.54%	95.54%
KUET			100.00%	100.00%
LUS			100.00%	100.00%
MBSTU			100.00%	100.00%
MIST			100.00%	100.00%
MIU			100.00%	100.00%
NINMAS			100.00%	100.00%
NSTU	580		98.70%	98.70%
NSU			100.00%	100.00%
NU			100.00%	100.00%
PSTU	174		99.61%	99.61%
PUST	281		99.37%	99.37%
RU			100.00%	100.00%
RUET			100.00%	100.00%
SAU			100.00%	100.00%
SBAU			100.00%	100.00%
SUST			100.00%	100.00%

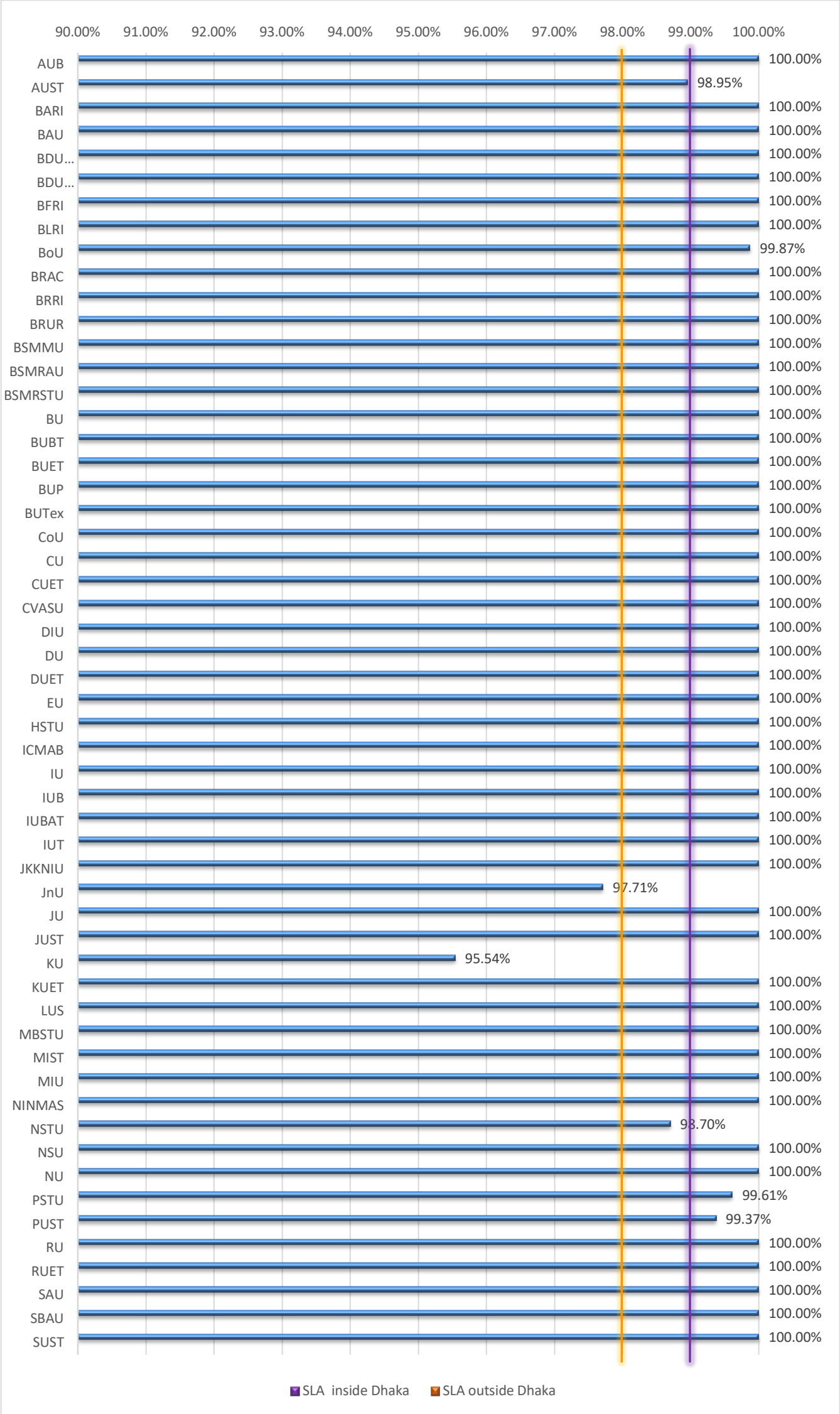


Figure 1: Institution Wise Link Availability considering “Planned Maintenance” as non-outage

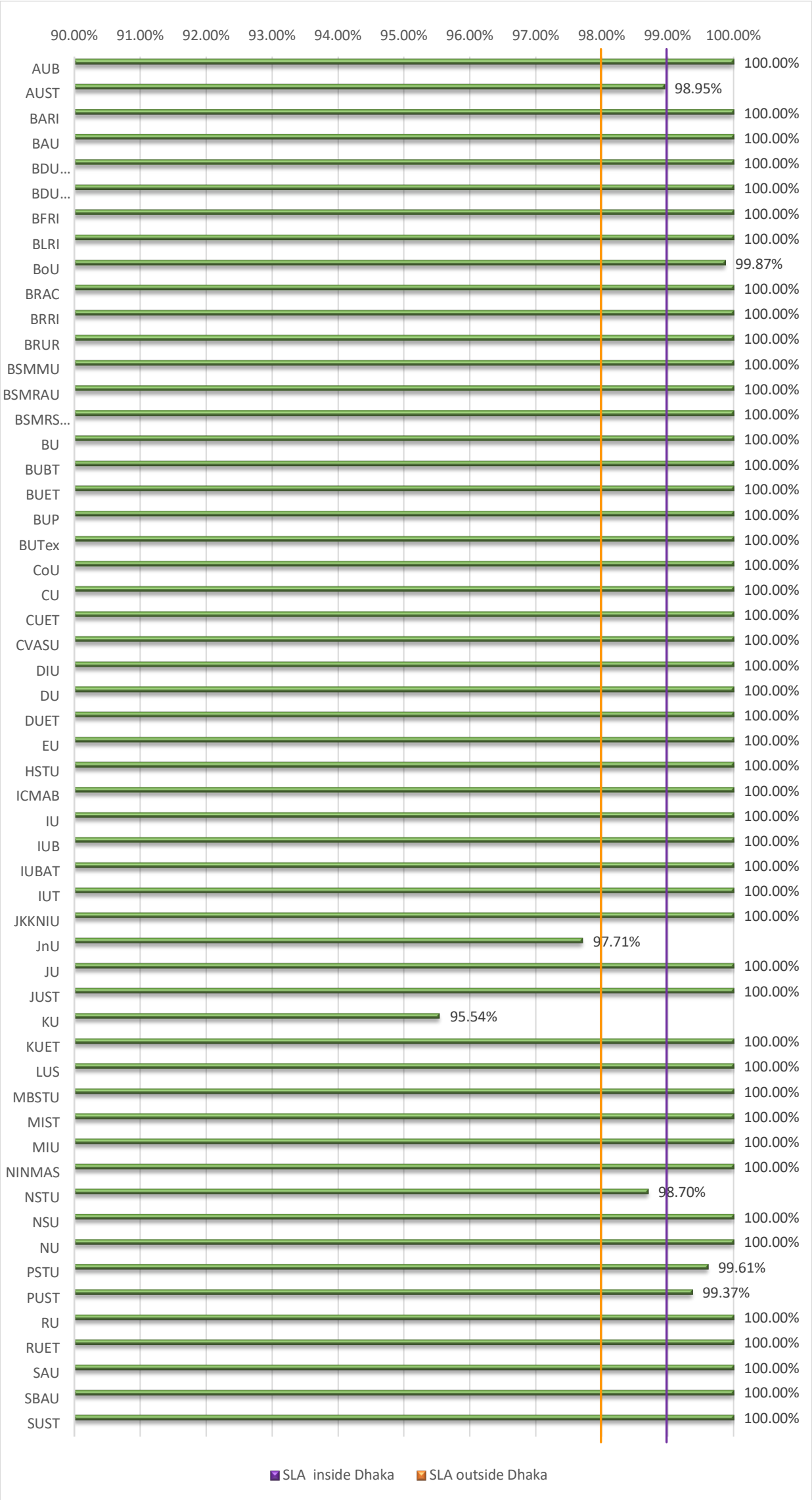


Figure 2: Institution Wise Link Availability considering “Planned Maintenance” as outage

3. MAJOR OUTAGE OF NOVEMBER 2021

- ❖ **Ticket#2021110601000025**
 - Affected Member** : PSTU
 - Recovery Time** : 154 min
 - SLA Compliance** : Fulfilled
 - Problem Details** : There was a problem on SMR-2 card in transmission device. The SMR-2 card did not work properly even after a hard reset. Later, BdREN team replaced the SMR-2 card and the link went up.

- ❖ **Ticket# 2021110701000014**
 - Affected Member** : AUST
 - Recovery Time** : 468 min
 - SLA Compliance** : Not Fulfilled
 - Problem Details** : While working near Shaheed Tajuddin Sarani, Tejgoan Industrial Area, a third-party company broke the BdREN fiber (U/G). The BdREN team was able to repair the broken fiber and bring the site back online.

- ❖ **Ticket# 2021111801000011, 2021111801000021**
 - Affected Member** : JNU
 - Recovery Time** : 1023 min
 - SLA Compliance** : Not Fulfilled
 - Problem Details** : Fiber was stolen near Fulbaria Market. BdREN along with HTL team had successfully laid 700m of o/h fiber and got the link up. Later, that day, there was fiber break behind Nagar Bhaban. BdREN team had successfully restored the broken fiber.

- ❖ **Ticket# 2021112401000018**
 - Affected Member** : KU
 - Recovery Time** : 1989 min
 - SLA Compliance** : Not Fulfilled
 - Problem Details** : A fiber break caused the link to go down. Our team was able to restore the fiber while also reducing the high-span loss. The link, however, did not go live. The BdREN team then took an SFP module into flight, replaced the faulty SFP and patch cord at the KUET-DIST end, and the link was restored.

- ❖ **Ticket# 2021120101000014**
 - Affected Member** : NSTU
 - Recovery Time** : 580 min
 - SLA Compliance** : Fulfilled
 - Problem Details** : There was high loss in transmission fiber about 6.5 KM away from NSTU. Mr. Kaswser (Technician) had successfully restored the broken fiber.

4. TRAFFIC STATISTICS

Table 2: Institution Wise Traffic Statistics

INSTITUTION	TOTAL TRAFFIC		AVERAGE BW		MAXIMUM BW	
	IN	OUT	IN	OUT	IN	OUT
DU	392.0 TB	52.0 TB	520.0 Mbps	80.0 Mbps	1400.0 Mbps	224.0 Mbps
BUET	374.0 TB	69.0 TB	496.0 Mbps	66.0 Mbps	963.0 Mbps	142.0 Mbps
BAU	354.0 TB	44.0 TB	224.0 Mbps	81.0 Mbps	892.0 Mbps	139.0 Mbps
CU	237.0 TB	35.0 TB	390.0 Mbps	50.0 Mbps	617.0 Mbps	112.0 Mbps
RU	232.0 TB	30.0 TB	372.0 Mbps	40.0 Mbps	646.0 Mbps	102.0 Mbps
CUET	183.0 TB	21.3 TB	487.0 Mbps	55.0 Mbps	900.7 Mbps	131.0 Mbps
IU	169.0 TB	13.0 TB	490.0 Mbps	38.0 Mbps	786.0 Mbps	95.0 Mbps
JU	165.0 TB	16.0 TB	232.0 Mbps	26.0 Mbps	548.0 Mbps	125.0 Mbps
RUET	159.0 TB	23.0 TB	288.0 Mbps	38.0 Mbps	561.0 Mbps	83.0 Mbps
DUET	119.0 TB	10.0 TB	226.0 Mbps	21.0 Mbps	444.0 Mbps	81.0 Mbps
JUST	105.0 TB	11.0 TB	163.0 Mbps	15.0 Mbps	362.0 Mbps	109.0 Mbps
CoU	101.0 TB	8.6 TB	125.0 Mbps	9.0 Mbps	224.0 Mbps	24.0 Mbps
CVASU	95.0 TB	11.3 TB	148.0 Mbps	13.3 Mbps	270.0 Mbps	33.0 Mbps
MBSTU	92.0 TB	10.0 TB	118.0 Mbps	12.0 Mbps	240.0 Mbps	36.0 Mbps
SBAU	90.0 TB	8.0 TB	132.0 Mbps	9.0 Mbps	372.0 Mbps	29.0 Mbps
BU	87.0 TB	6.5 TB	121.0 Mbps	8.0 Mbps	227.0 Mbps	43.1 Mbps
PSTU	85.0 TB	10.0 TB	369.0 Mbps	42.0 Mbps	677.0 Mbps	143.0 Mbps
BRUR	73.0 TB	6.0 TB	110.0 Mbps	23.0 Mbps	303.0 Mbps	88.0 Mbps
BSMMU	63.0 TB	6.0 TB	91.0 Mbps	7.0 Mbps	303.0 Mbps	37.0 Mbps
BUP	47.0 TB	10.0 TB	35.6 Mbps	8.0 Mbps	163.0 Mbps	82.5 Mbps
JKKNIU	45.6 TB	3.8 TB	62.0 Mbps	4.0 Mbps	203.0 Mbps	18.0 Mbps
BOU	39.0 TB	18.0 TB	75.0 Mbps	26.0 Mbps	288.0 Mbps	72.0 Mbps
NSTU	35.6 TB	3.3 TB	39.3 Mbps	4.0 Mbps	89.0 Mbps	34.0 Mbps
JnU	35.0 TB	5.0 TB	59.0 Mbps	11.0 Mbps	237.0 Mbps	55.0 Mbps
SUST	33.0 TB	4.0 TB	100.0 Mbps	14.0 Mbps	268.0 Mbps	69.0 Mbps
IUT	33.0 TB	7.0 TB	175.0 Mbps	37.0 Mbps	490.0 Mbps	127.0 Mbps
BSMRAU	33.0 TB	4.0 TB	55.0 Mbps	19.0 Mbps	126.0 Mbps	47.0 Mbps
KUET	22.0 TB	3.0 TB	100.0 Mbps	9.0 Mbps	244.0 Mbps	33.0 Mbps
IUB	21.0 TB	2.7 TB	34.0 Mbps	4.0 Mbps	107.0 Mbps	54.0 Mbps
BUTex	20.0 TB	2.0 TB	45.0 Mbps	4.0 Mbps	157.0 Mbps	94.0 Mbps
NU	20.0 TB	2.4 TB	9.1 Mbps	0.6 Mbps	55.5 Mbps	16.0 Mbps
BSMRSTU	17.0 TB	1.5 TB	62.0 Mbps	5.0 Mbps	120.0 Mbps	14.0 Mbps
PUST	14.5 TB	1.2 TB	81.0 Mbps	7.0 Mbps	181.0 Mbps	18.0 Mbps
KU	12.3 TB	1.4 TB	23.0 Mbps	2.0 Mbps	68.0 Mbps	18.0 Mbps
HSTU	12.1 TB	1.3 TB	9.5 Mbps	1.1 Mbps	48.0 Mbps	15.0 Mbps
SAU	11.0 TB	0.8 TB	42.0 Mbps	2.4 Mbps	116.0 Mbps	8.0 Mbps
DIU	11.0 TB	104.0 TB	53.0 Mbps	392.0 Mbps	198.0 Mbps	733.0 Mbps
BDU	10.0 TB	1.8 TB	12.0 Mbps	5.0 Mbps	65.0 Mbps	17.0 Mbps
BLRI	9.5 TB	0.9 TB	29.4 Mbps	2.8 Mbps	73.0 Mbps	16.0 Mbps
BARI	7.7 TB	0.8 TB	24.0 Mbps	2.4 Mbps	155.0 Mbps	67.0 Mbps
BRRI	7.4 TB	0.7 TB	23.0 Mbps	2.3 Mbps	115.0 Mbps	48.0 Mbps
MIST	4.0 TB	5.4 TB	10.0 Mbps	4.0 Mbps	40.0 Mbps	18.0 Mbps
EU	3.5 TB	0.5 TB	20.0 Mbps	2.7 Mbps	113.0 Mbps	13.0 Mbps
BFRI	3.5 TB	0.3 TB	4.5 Mbps	0.5 Mbps	64.0 Mbps	15.0 Mbps
AUST	3.3 TB	0.4 TB	2.6 Mbps	0.2 Mbps	30.0 Mbps	1.0 Mbps
LU	3.1 TB	1.0 TB	22.0 Mbps	5.5 Mbps	101.0 Mbps	78.0 Mbps
BRAC	3.0 TB	1.0 TB	15.0 Mbps	4.0 Mbps	122.0 Mbps	76.0 Mbps
ICMAB	2.0 TB	1.3 TB	2.0 Mbps	0.8 Mbps	87.0 Mbps	61.0 Mbps
AUB	1.0 TB	0.9 TB	5.0 Mbps	1.0 Mbps	43.0 Mbps	38.0 Mbps
NSU	0.0 TB	0.0 TB	0.0 Mbps	0.0 Mbps	0.0 Mbps	0.0 Mbps
VU	0.0 TB	0.0 TB	0.0 Mbps	0.0 Mbps	0.0 Mbps	0.0 Mbps
EWU	0.0 TB	0.0 TB	0.0 Mbps	0.0 Mbps	0.0 Mbps	0.0 Mbps

5. TOTAL TRAFFIC

Table 3: Total Traffic

Total Download	3695.1 TB
Total Upload	581.14 TB

6. ONLINE EDUCATION

6.1 Institution Wise vSession Usage Statistics

➤ Public Universities and Colleges:

Table 4: vSession Usage by Public Universities and Colleges

SL	University_name	Total Class Taken	Total Hours	Total Participants
1	Jahangirnagar University (JU)	1471	2201	62749
2	Begum Rokeya University, Rangpur (BRUR)	876	1039	51475
3	Jagannath University (JnU)	788	935	30531
4	University of Dhaka (DU)	753	1037	25516
5	Noakhali Science and Technology University (NSTU)	733	908	31798
6	Hajee Mohammad Danesh Science & Technology University (HSTU)	669	763	48419
7	Lalmatia Mohila College	623	370	8166
8	University of Chittagong (CU)	614	851	28046
9	Bangladesh Agricultural University (BAU)	570	639	20678
10	University of Rajshahi (RU)	415	471	11386
11	Islamic University of Technology (IUT)	381	589	10847
12	Military Institute of Science and Technology (MIST)	375	446	21622
13	Comilla University (CoU)	354	386	12936
14	Khulna University (KU)	313	366	10913
15	Bangladesh University of Textiles (BUTEX)	308	370	16835
16	National Institute of Textile Engineering and Research (NITER)	278	293	26729
17	Mawlana Bhashani Science and Technology University (MBSTU)	248	330	7252
18	University of Barisal (BU)	211	261	10887
19	Bangladesh Open University (BOU)	198	282	5304
20	BSMRSTU	197	227	9078
21	Bangladesh University of Professionals (BUP)	188	248	10424
22	Jashore University of Science and Technology (JUST)	187	175	4575
23	Eden Mohila College	145	131	8770
24	Cumilla Shikkha Board Govt. Model College	112	50	2500
25	Sher-e-Bangla Agricultural University (SBAU)	106	109	3791
26	Jatiya Kabi Kazi Nazrul Islam University (JKKNIU)	104	129	4109
27	Islamic University, Bangladesh (IU)	101	105	4410
28	Bangabandhu Sheikh Mujib Medical University (BSMMU)	99	132	2542
29	Bangladesh University of Engineering & Technology (BUET)	99	142	5902
30	Bangabandhu Sheikh Mujibur Rahman Maritime University	80	107	2185
31	Pabna University of Science and Technology (PUST)	75	93	3141
32	Chittagong Veterinary and Animal Sciences University (CVASU)	75	88	3324
33	Patuakhali Science and Technology University (PSTU)	71	102	1669
34	Shahjalal University of Science and Technology (SUST)	60	82	1603
35	Chittagong University of Engineering & Technology (CUET)	53	94	2248
36	National University (NU)	52	53	1153
37	Dhaka University of Engineering & Technology (DUET)	50	58	1261
38	Comilla Victoria Government College	45	42	703
39	Sylhet Agricultural University (SAU)	40	78	1497

40	Rangamati Science and Technology University (RMSTU)	36	39	1194
41	Khulna Agricultural University (KAU)	35	50	199
42	Bangabandhu Sheikh Mujibur Rahman Agricultural University	32	36	403
43	Mohammadpur Govt college, Dhaka	29	43	249
44	BSFMSTU	18	22	358
45	Shahid Abdur Rab Serniabat Textile Engineering College (SARSTEC)	16	31	869
46	Rajshahi University of Engineering & Technology (RUET)	13	12	1027
47	Textile Engineering College, Chattogram (TEC)	13	20	1474
48	Rabindra University, Bangladesh (RUB)	11	9	440
49	Khulna University of Engineering & Technology (KUET)	10	8	290
50	Sheikh Kamal Textile Engineering College, Jhenaidah (SKTEC)	8	4	297
51	Bangabandhu Digital University (BDU)	8	12	83
52	Chandpur Government College, Chandpur	8	6	456
53	Sheikh Hasina University (SHU)	6	6	185
54	Faridpur Engineering College, Faridpur	5	5	93
55	Mymensingh Engineering College	4	5	57
56	BSMRAAU	2	2	7
57	Pabna Textile Engineering College (PTEC)	1	2	99

➤ **Private Universities:**

Table 5: vSession Usage by Private University

SL	Institution Name	Total Class	Total Hours	Total Participants
1	International Islamic University Chittagong (IIUC)	6259	7242	213916
2	Green University of Bangladesh	5578	7477	227260
3	Stamford University Bangladesh	3604	4959	106322
4	Shanto-Mariam University of Creative Technology	2889	4099	79892
5	Primeasia University	2752	3105	64856
6	Bangladesh Army International University of Science and Tech	1610	1684	45504
7	Premier University, Chittagong	1539	2002	66761
8	BGC Trust University Bangladesh	1466	1524	35648
9	BGMEA University of Fashion & Technology	1260	1734	51864
10	Ahsanullah University of Science and Technology (AUST)	1253	2327	42666
11	Eastern University	1233	1612	43911
12	City University	1115	1436	44906
13	ASA University Bangladesh	902	1252	12864
14	University of Science and Technology Chittagong (USTC)	821	757	18325
15	Presidency University	803	928	21148
16	State University of Bangladesh	649	849	22815
17	Canadian University of Bangladesh	626	594	8462
18	BRAC University	587	840	22919
19	Northern University of Business & Technology Khulna	578	698	20812
20	Manarat International University	433	547	15882
21	Leading University, Sylhet	378	447	8885
22	Rajshahi Cantonment Public School and College (RCPSC)	373	267	19827
23	University of Asia Pacific	346	587	13451
24	Dhaka International University (DIU)	346	387	10299
25	Notre Dame University Bangladesh	329	402	7363
26	CCN University of Science & Technology, Cumilla	319	311	6464
27	Southeast University	308	418	11769
28	University of South Asia	308	382	12241
29	Bangladesh University of Health Sciences	263	335	5942
30	Sonargaon University (SU)	236	249	10286
31	Asian University of Bangladesh (AUB)	177	205	2755
32	University of Liberal Arts Bangladesh (ULAB)	175	298	7418
33	North Western University (Khulna)	157	182	2113

SL	Institution Name	Total Class	Total Hours	Total Participants
34	University of Information Technology and Sciences (UITS)	136	216	4306
35	Gono Bishwabidyalay	135	138	3702
36	Bangladesh International School & College (BISC)	131	156	3553
37	Atish Dipankar University of Science and Technology	92	117	10130
38	International University of Business Agriculture and Technology	90	122	4486
39	Port City International University	83	85	3660
40	North East University Bangladesh	81	88	2190
41	Bangladesh Army University of Science and Technology (BAUST)	80	86	1999
42	Prime University	80	107	1952
43	European University of Bangladesh	79	84	1777
44	Znrf University of Management science	60	104	756
45	East West University (EWU)	59	83	2115
46	Feni University	55	58	947
47	Rabindra Maitree University	55	66	969
48	International Standard University (ISU)	53	67	1175
49	Adamjee Cantonment College	51	48	445
50	Bangladesh Islami University (BIU)	50	77	943
51	North South University	44	51	511
52	United International University (UIU)	40	79	995
53	Chittagong Independent University (CIU)	38	44	974
54	Bangladesh Army University of Engineering and Technology	26	28	809
55	Pundra University of Science & Technology	23	25	345
56	Varendra University	15	20	820
57	Exim Bank Agricultural University Bangladesh	15	13	128
58	The International University of Scholars	15	19	531
59	Bangladesh University	14	22	117
60	East Delta University	14	19	366
61	Daffodil International University	9	11	1341
62	University of Creative Technology Chittagong (UCTC)	6	9	77
63	Notre Dame College, Dhaka	4	5	92
64	Fareast International University (FIU)	1	3	28
65	Savar Cantonment Public School & College (SCPSC)	1	1	97
66	Ranada Prasad Shaha University (RPSU)	1	3	4

➤ **Medical Colleges:**

Table 6: vSession Usage by Medical Colleges

SL	Institution Name	Total Class	Total Hours	Total Participants
1	Ibrahim Medical College, Dhaka	154	229.2165	8489
2	TMSS Medical College	130	148.2004	10809
3	Holy Family Red Crescent Medical College	91	86.0499	3317
4	Dhaka Medical College	58	70.8499	6320
5	Faridpur Medical College	52	52.9336	2863
6	North East Medical College (NEMC)	17	14.5169	1530
7	East-West Medical College & Hospital (EWMCH)	8	6.4001	227
8	MAG Osmani Medical College, Sylhet	3	2.2166	321
9	Sher-e-Bangla Medical College, Barisal	3	3.4834	333
10	US-Bangla Medical College	3	1.4167	153
11	Rajshahi Medical University	1	0.8	14

➤ **Other Institution:**

Table 7: vSession Usage by Other Institution

SL	Institution Name	Total Class	Total Hours	Total Participants
1	Bangladesh Institute of Management (BIM)	78	154.367	5086
2	Bangladesh Institute of Capital Market (BICM)	47	82.1668	792
3	Institute of Chartered Secretaries of Bangladesh (ICSB)	32	66.3167	1060
4	Government Organization	19	27.3499	641
5	Sylhet Engineering College (SEC)	19	25.8167	383
6	Shaikh Burhanuddin Post Graduate College	10	7.0165	371
7	Bangladesh Academy for Securities Markets - BASM	8	11.9168	44
8	National Institute of Population Research & Training	8	21.95	228
9	Ministry of Education	7	12.6333	507
10	BANBEIS	7	23.5	568
11	Bangladesh Institute of Nuclear Agriculture (BINA)	4	6.0667	163
12	Mymensing Girls' Cadet College	2	3.6167	138
13	Bangladesh Research and Education Network (BdREN)	1	1.0167	9
14	Bangladesh Association of Women Scientist (BAWS)	1	1.7	38
15	Technical Teachers' Training College	1	1.25	10

6.2 Institution Wise vSession Usage Graph

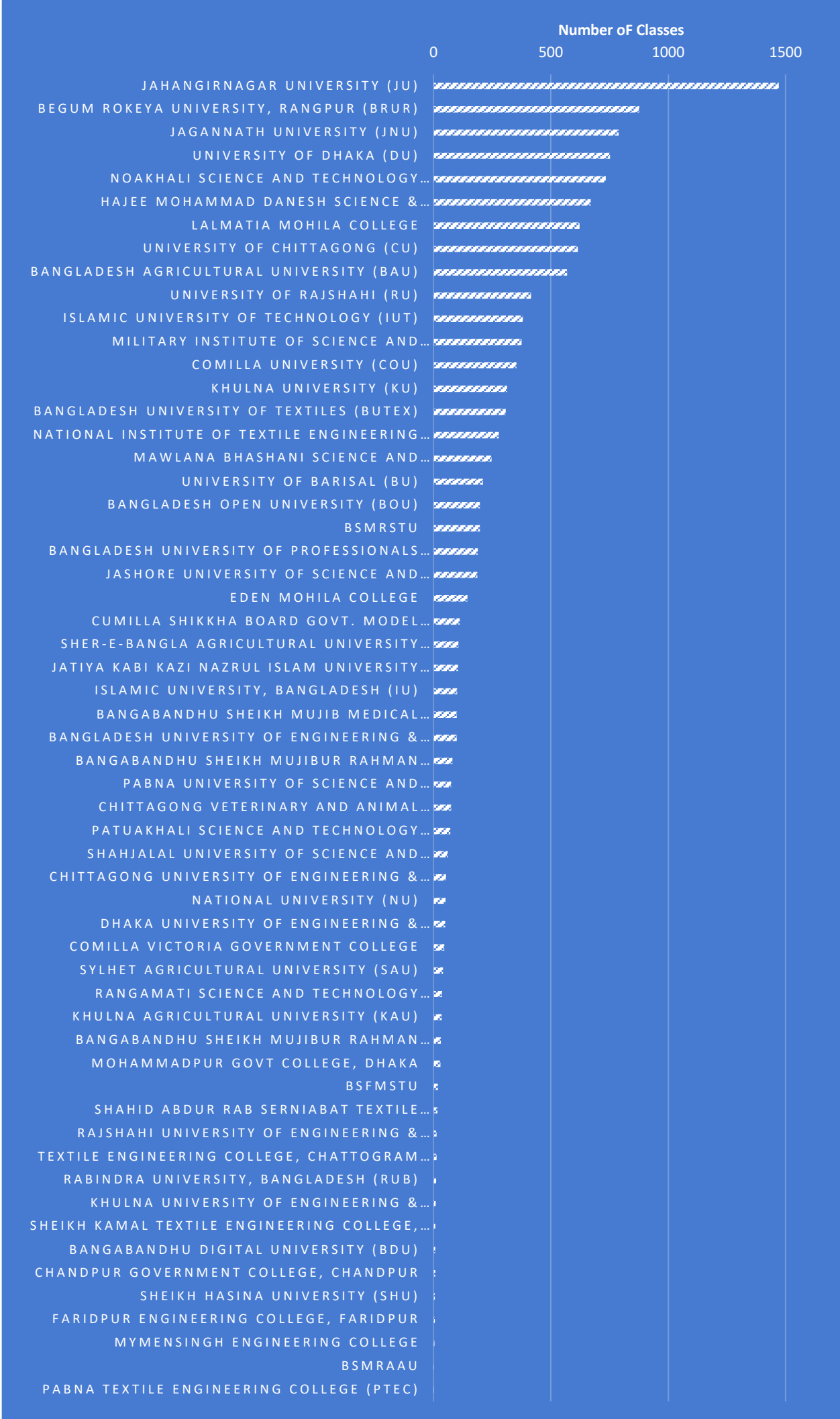


Figure 3: Public University & Colleges vSession Usage Graph

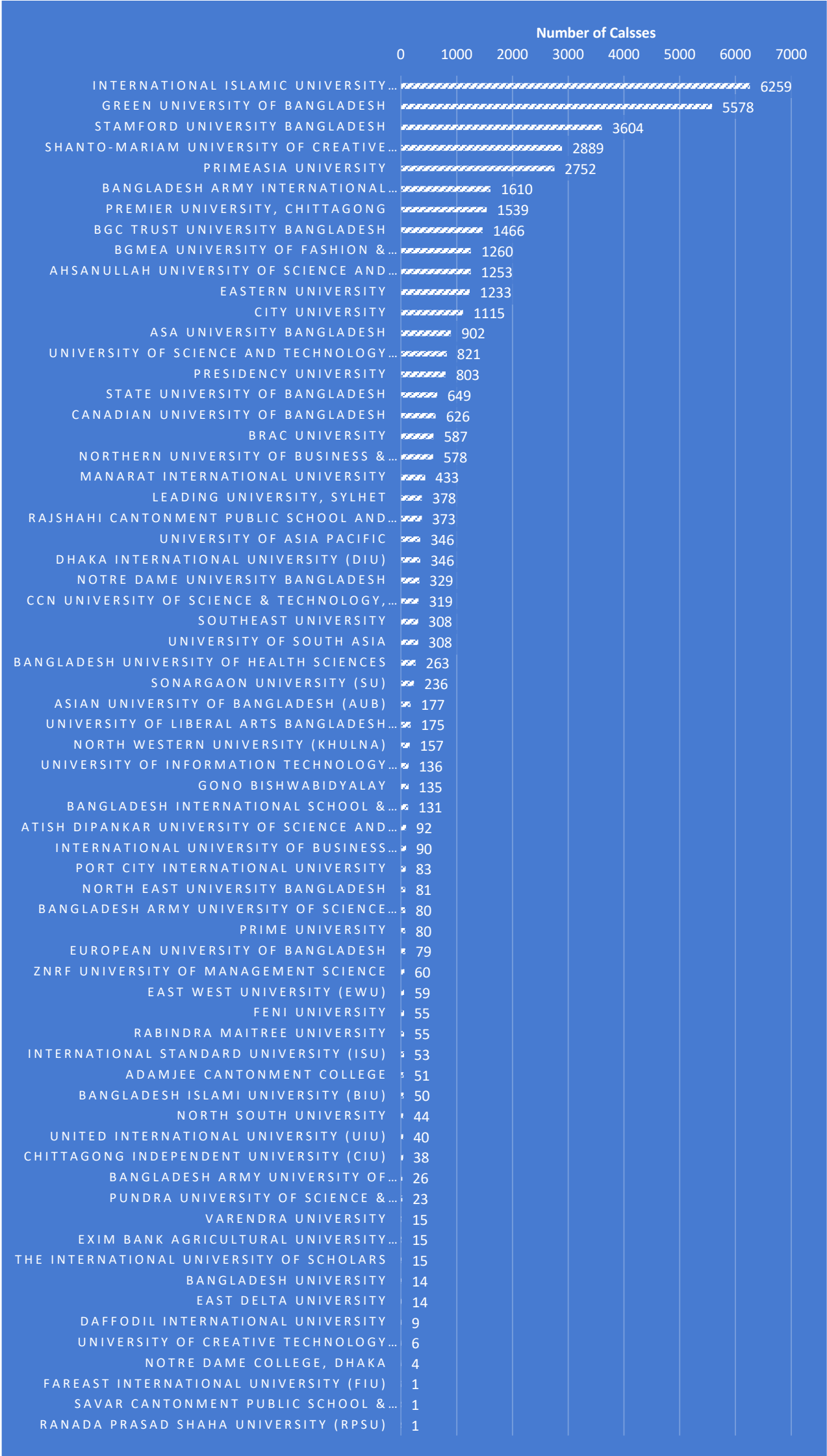


Figure 4: Private University vSession Usage Graph

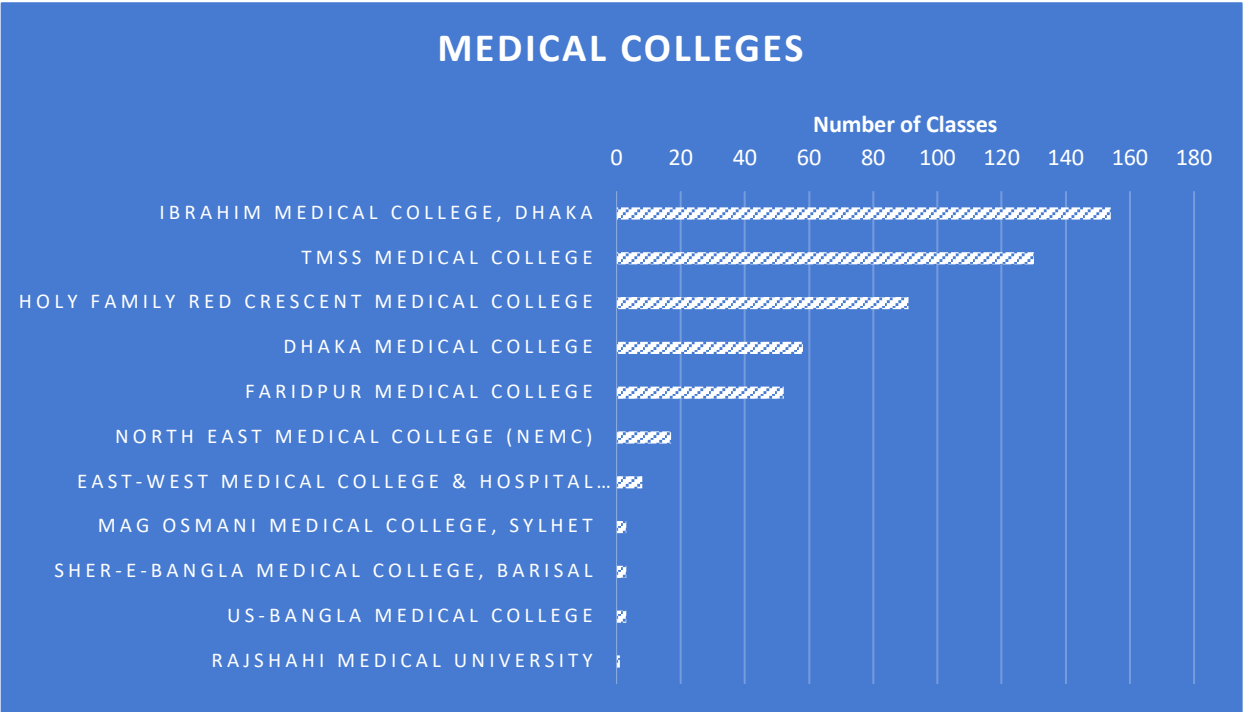


Figure 5: Medical College vSession Usage Graph

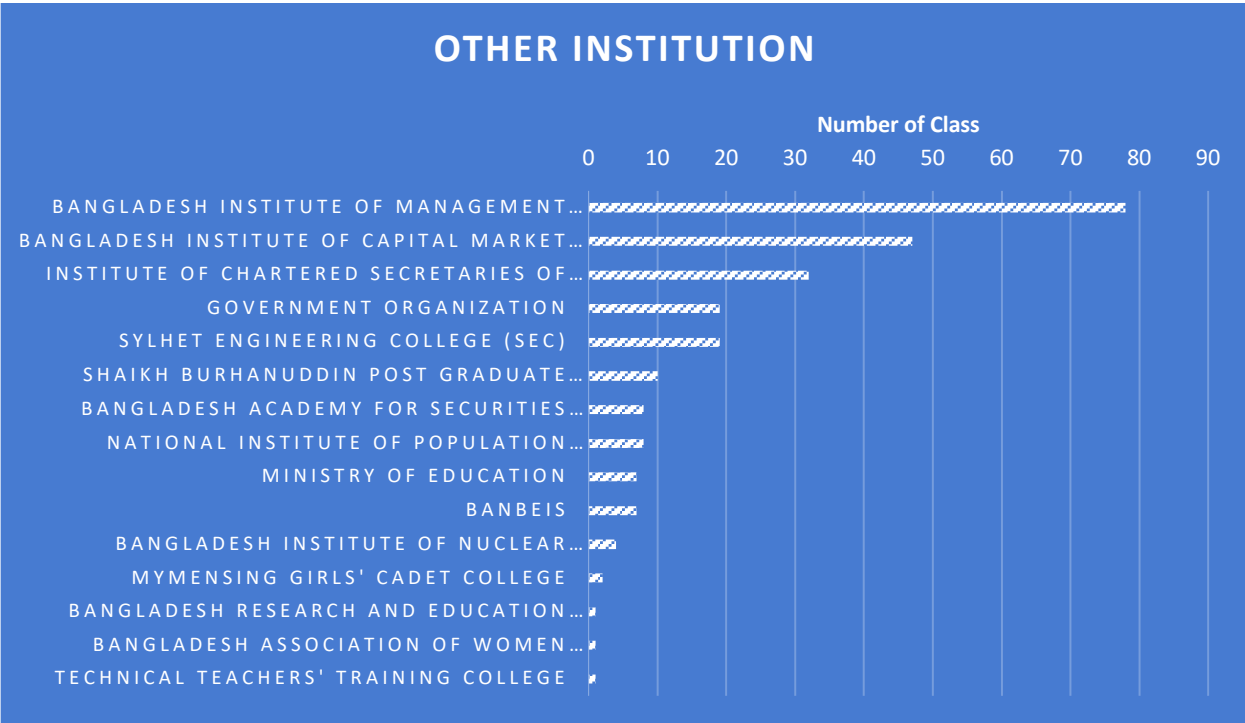


Figure 6: Other Institution vSession Usage Graph

6.3 Institution Wise On-Prem Zoom Usage Statistics

Table 8:Institution Wise On-Prem Zoom Usage Statistics

Sl.	Name of The Institution	Number of Class	Sum of Duration (hh:mm:ss)	Sum of Participants
1	University of Dhaka (DU)	3762	5441:56:00	202031
2	Bangladesh University of Engineering & Technology (BUET)	2468	3227:25:47	58659
3	Bangladesh University of Professionals (BUP)	2330	3322:03:11	224390
4	Military Institute of Science and Technology (MIST)	1876	2321:40:04	121291
5	Khulna University of Engineering & Technology (KUET)	943	1595:15:52	44599
6	Shahjalal University of Science and Technology (SUST)	923	1510:36:21	42437

Sl.	Name of The Institution	Number of Class	Sum of Duration (hh:mm:ss)	Sum of Participants
7	Chittagong University of Engineering & Technology (CUET)	864	1422:04:38	45560
8	University of Dhaka	854	1275:27:10	63246
9	United International University (UIU)	750	1043:38:53	25821
10	Dhaka University of Engineering & Technology (DUET)	640	1139:04:58	33458
11	Khulna University (KU)	568	963:01:28	23777
12	Jahangirnagar University (JU)	429	719:04:08	22469
13	Islamic University of Technology (IUT)	398	527:06:58	14620
14	University of Chittagong (CU)	392	535:26:02	18183
15	Rajshahi University of Engineering & Technology (RUET)	284	305:58:51	6035
16	ICMAB	255	426:09:47	22143
17	Jashore University of Science and Technology (JUST)	232	271:59:18	8193
18	Sylhet Agricultural University (SAU))	218	388:48:57	12656
19	Bangladesh Agricultural University (BAU)	195	221:12:04	7117
20	Kumudini Women's Medical College	163	494:01:21	30927
21	Noakhali Science and Technology University (NSTU)	142	180:49:47	7331
22	Patuakhali Science and Technology University (PSTU)	137	211:56:57	6575
23	Bangladesh Open University (BOU)	125	146:40:30	4615
24	University of Rajshahi (RU)	125	145:47:49	4543
25	Pabna University of Science and Technology (PUST)	121	161:01:00	7118
26	Jagannath University (JnU)	109	134:36:36	3469
27	Manarat International University	109	143:59:53	5115
28	Hajee Mohammad Danesh Science & Technology University (HSTU)	97	146:49:22	9165
29	National University (NU)	97	206:32:39	7132
30	BdREN	86	104:42:20	1012
31	Joypurhat Girls Cadet College	86	65:38:07	9787
32	Rajshahi Cantonment Public School and College (RCPSC)	78	59:53:05	3963
33	Asian University of Bangladesh (AUB)	74	109:13:45	2820
34	University of Barisal (BU)	74	85:41:08	3496
35	Green University of Bangladesh	73	96:47:48	3003
36	Begum Rokeya University, Rangpur (BRUR)	62	73:59:35	4670
37	Rajshahi Cadet College	62	65:30:17	4757
38	Chittagong Medical College	54	226:36:18	11951
39	International Islamic University Chittagong (IIUC)	54	68:05:47	1706
40	Jhenidah Cadet College (JCC)	52	135:17:35	6911
41	UGC	51	102:23:39	2303
42	Comilla University (CoU)	48	59:03:06	3309
43	BARD	46	84:58:57	3404
44	Canadian University of Bangladesh	45	58:41:58	2066
45	Jatiya Kabi Kazi Nazrul Islam University (JKKNIU)	44	68:35:18	1867
46	BSMRSTU	43	46:39:51	1864
47	Mawlana Bhashani Science and Technology University (MBSTU)	43	67:31:04	2301

Sl.	Name of The Institution	Number of Class	Sum of Duration (hh:mm:ss)	Sum of Participants
48	BSMRAAU	41	71:09:05	591
49	BGMEA University of Fashion & Technology	40	83:39:37	3307
50	SOMC	40	82:42:44	8986
51	Bangladesh International School & College (BISC)	39	34:45:43	2632
52	MH Samorita Medical College	39	66:16:59	5343
53	IEDCR	37	43:19:38	659
54	Spectrum Engineering Consortium Limited	37	50:27:08	404
55	SPFMS	36	74:52:25	837
56	Data Edge	35	53:50:40	645
57	Ministry of Education	35	69:26:47	1606
58	IUT Temporary	34	75:46:26	1541
59	International Standard University (ISU)	33	46:10:18	1056
60	BGC Trust University Bangladesh	31	29:18:26	567
61	ICDDRDB	31	44:15:34	1316
62	Local Government Engineering Department(LGED)	31	78:09:21	213
63	University of Liberal Arts Bangladesh (ULAB)	31	64:30:15	5820
64	Ahsanullah University of Science and Technology (AUST)	30	30:34:37	929
65	Bangladesh Agricultural Research Institute (BARI)	30	87:08:25	4156
66	Stamford University Bangladesh	30	48:34:15	546
67	East West University (EWU)	29	53:09:47	1272
68	Institute of Chartered Secretaries of Bangladesh (ICSB)	29	40:04:59	967
69	Sher-e-Bangla Agricultural University (SBAU)	29	33:06:29	2623
70	Gono Bishwabidyalay	28	51:02:09	1501
71	BTCL	26	60:11:19	1792
72	City University	26	40:19:52	1639
73	Presidency University	26	37:39:43	530
74	Bangabandhu Sheikh Mujibur Rahman Agricultural University	25	33:47:16	1475
75	NIPORT	25	32:37:01	1706
76	Bangladesh Institute of Capital Market (BICM)	24	51:50:47	916
77	Bangladesh Institute of Management (BIM)	24	28:53:34	70
78	Daffodil International University	24	32:33:38	1309
79	Rangamati Science and Technology University (RMSTU)	24	54:19:38	307
80	Savar Cantonment Public School & College (SCPSC)	24	16:15:32	1206
81	BPC Training Center	22	44:25:17	1137
82	Chittagong Independent University (CIU)	21	22:23:12	935
83	University of Science and Technology Chittagong (USTC)	21	13:50:13	253
84	Chittagong Veterinary and Animal Sciences University (CVASU)	20	23:48:53	688
85	Sylhet Cadet College (SCC)	20	23:23:24	1794
86	Faridpur Engineering College, Faridpur	19	42:26:54	5270

Sl.	Name of The Institution	Number of Class	Sum of Duration (hh:mm:ss)	Sum of Participants
87	fDLuDCf	19	17:45:03	279
88	Bangladesh Sugar & Food Industries Corporation (BSFIC)	18	73:10:38	2579
89	Bangladesh University of Textiles (BUTEX)	18	22:57:44	580
90	IVTI	18	21:32:59	370
91	Adamjee Cantonment College	17	19:46:59	88
92	Bangabandhu Digital University (BDU)	17	25:16:06	1138
93	Customs Intelligence & Investigation Directorate	17	40:21:11	1602
94	Bangladesh Rice Research Institute (BRRI)	16	50:31:51	855
95	Rangpur Medical College (RMC)	14	30:23:28	1956
96	Sher-e-Bangla Medical College, Barisal	14	18:35:13	1759
97	Barishal Cadet College (BCC)	13	15:14:36	223
98	Bangladesh Institute of Development Studies	12	21:15:33	288
99	Bangladesh Securities & Exchange Commission (BSEC)	12	24:55:04	1485
100	Independent University, Bangladesh	11	17:36:40	504
101	North Western University (Khulna)	11	11:47:56	190
102	Port City international University	11	16:37:05	659
103	Dhamrai Govt. College	10	6:14:01	363
104	BRAC University	8	9:03:36	128
105	Islamic University, Bangladesh (IU)	8	8:39:07	163
106	Rajshahi Medical University	8	5:40:41	406
107	North South University	7	11:31:09	221
108	Faith Bangladesh	6	6:05:03	31
109	Govt. Office	6	5:44:11	125
110	Kazla Technologies Limited	6	2:50:50	30
111	Notre Dame College, Dhaka	6	12:10:44	1742
112	Dhaka Dental College	5	14:46:54	735
113	NIO	5	6:47:52	713
114	University of Asia Pacific	5	7:33:54	97
115	BAC	4	11:36:00	1216
116	Bangabandhu Sheikh Mujib Medical University (BSMMU)	4	2:18:07	34
117	Shaikh Burhanuddin Post Graduate College	4	2:30:21	65
118	Dhaka Medical College	3	3:27:28	993
119	Govt. Ashek Mahmud College	3	3:30:58	58
120	Holy Family Red Crescent Medical College	3	2:37:16	435
121	International University of Business Agriculture and Technology	3	9:13:43	459
122	ABC Center for HR Development	2	6:39:56	26
123	Anwer Khan Modern University	2	3:15:51	7
124	Bangladesh Livestock Research Institute (BLRI)	2	4:24:46	58
125	Bangladesh University of Business and Technology (BUBT)	2	3:43:19	37
126	Department of Telecommunications	2	1:32:36	8
127	Dr. M.A Wazed Miah Textile Engineering College	2	1:37:15	65
128	Textile Engineering College, Noakhali	2	1:06:42	16
129	Cumilla Victoria Government College	1	2:24:32	76
130	Dhaka International University (DIU)	1	0:12:23	3

Sl.	Name of The Institution	Number of Class	Sum of Duration (hh:mm:ss)	Sum of Participants
131	EXIM Bank Agricultural University Bangladesh	1	1:56:38	102
132	Northern University of Business & Technology Khulna	1	4:01:40	753
133	Ranada Prasad Shaha University (RPSU)	1	1:18:17	5
134	Sheikh Hasina University (SHU)	1	1:29:00	65

6.4 BdREN Total Zoom Usage Statistics

Table 9: BdREN Total Zoom Usage Statistics

Total Number of Meetings/Classes	76633
Total Hour of Meetings/Classes	101018:36:00
Total Number of Participants	3153253

6.5 Month-wise Number of Classes, Total Class Hours and Participants

Table 10: Month-wise Zoom Usage Statistics

Year 2021	Number Of Classes	Total Class Hours	Total Participants
January	88839	111686	4896004
February	144422	1,73,143	7082885
March	111179	135669	3400250
April	115737	139956	5834405
May	79420	102116	3215695
June	127770	166413	6441336
July	111348	149652	5378990
August	147295	202190	7435008
September	133279	179272	6365539
October	91010	116065	3653814
November	76633	101018	3153253

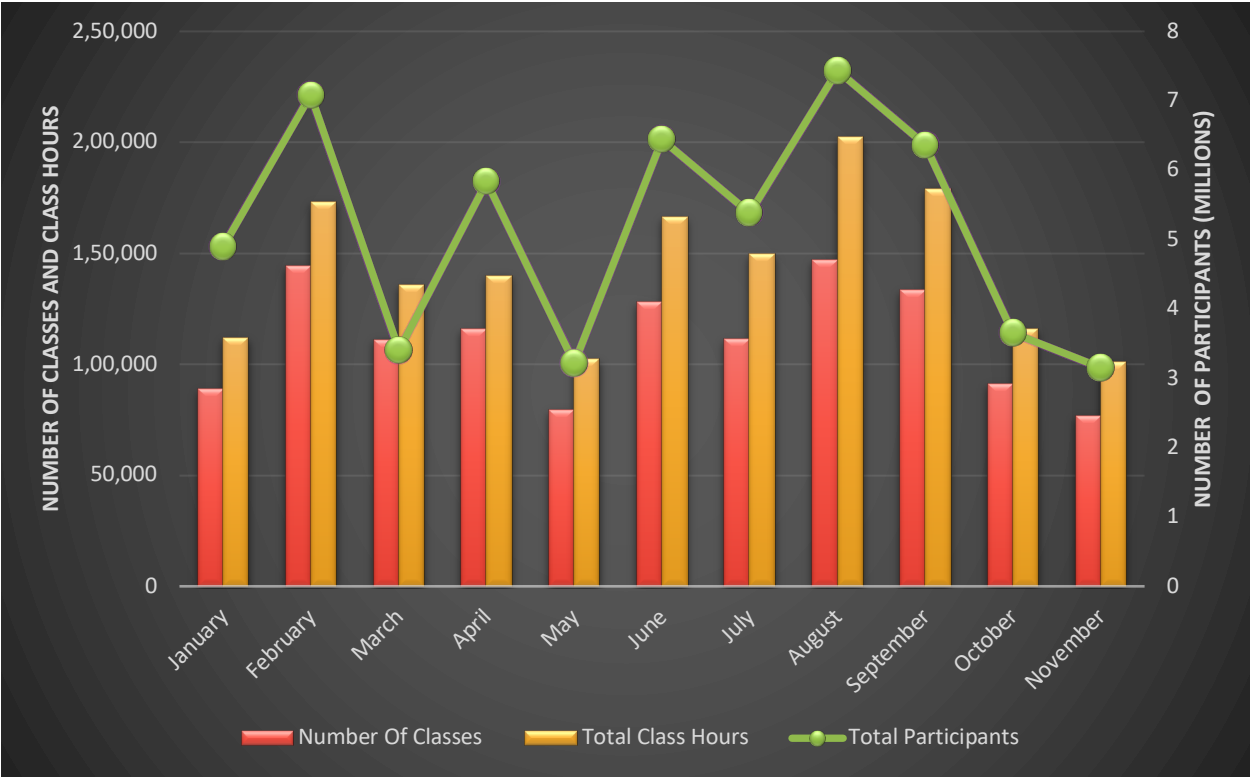


Figure 7: Month Wise Zoom Usage Graph in 2021

7. INSTITUTION WISE MAX HPC USAGE

Table 11: Institution Wise Max HPC Usage

Institutions	Used Space (GB)	Memory Size (GB)	CPUs
Institutions	6904.2	468	104
BAU	3.17	4	2
BRAC	15.2	12	4
BRACU	1024	24	8
BRU	1024	8	32
BSMMU	8.11	8	4
BSMRAAU	4602.11	168	64
BSMRAU	150.75	96	48
BU	52.9	17	11
BUET	80.33	80	20
BUP	480.33	16	8
CEDP	250	16	8
CU	41.08	32	16
CUET	9499.37	104	36
CVASU	12174.7	2083.96	578
DU	100.02	48	48
DUET	149.79	64	32
EWU	16.11	16	8
HSTU	1024.22	24	12
IAU	1024.22	96	32
IndU	178.45	76	24
IU	3.86	4	4
IRS	11.64	8	2
IUT	12.2	4	4
IUS	32.11	32	16
JNU	2030	32	12
JKKNIU	1492.98	72	44
JU	2607.42	124	60
JUST	40	40	28
KU	56.76	32	20
KUET	2306.13	136	52
MBSTU	1010	8	4
NITER	20	32	16
NSU	1010	16	8
PSTU	1095.07	40	12
PUST	73.49	22	14
RUET	98.25	40	20
SAU	763.23	184	46
SUST	643.52	100	33
UGC	6904.2	468	104

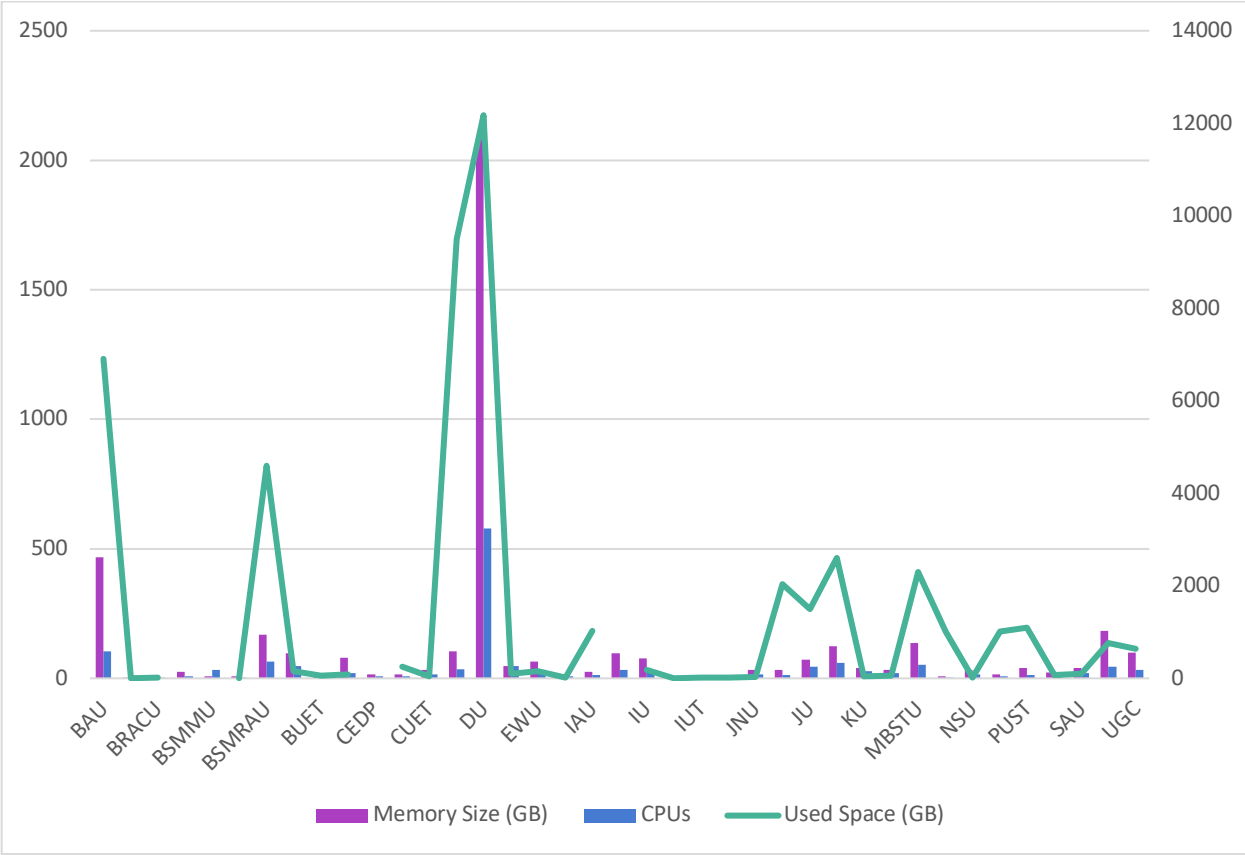


Figure 8: Institution Wise Max HPS Usage

8. ACRONYMS

AUB	Asian University of Bangladesh	JUST	Jessore University of Science & Technology
AUST	Ahsanullah University of Science & Technology	KU	Khulna University
AUW	Asian University for Women	KUET	Khulna University of Engineering & Technology Leading University, Sylhet
BARD	Bangladesh Academy for Rural Development	LUS	
BARI	Bangladesh Agricultural Research Institute	MBSTU	Mawlana Bhashani Science and Technology University
BAU	Bangladesh Agricultural University	MIU	Manarat International University
BDU	Bangabandhu Sheikh Mujibur Rahman Digital University	MMC	Mymensingh Medical College, Mymensingh
BdREN	Bangladesh Research and Education Network	NINMAS	National Institute of Nuclear Medicine & Allied Science
BFRI	Bangladesh Fisheries Research Institute	NITER	National Institute of Textile Engineering & Research
BIDS	The Bangladesh Institute of Development Studies	NKN	National Knowledge Network
BLRI	Bangladesh Livestock Research Institute	NOC	Network Operation Center
BOU	Banagladesh Open University	NREN	National Research and Education Network
BRACU	BRAC University	NSTU	Noakhali Science and Technology University
		NSU	North South University
BRRI	Bangladesh Rice Research Institute	NU	National University
BRUR	Begum Rokeya University, Rangpur	PSTU	Patuakhali Science and Technology University
BSMMU	Bangabandhu Sheikh Mujib Medical University	PUST	Pabna University of Science and Technology
BSMRAU	Bangabandhu Sheikh Mujibur Rahman Agricultural University	RAM	Random Access Memory
BSMRSTU	Bangabandhu Sheikh Mujibur Rahman Science and Technology University	RMC	Rajshahi Medical College, Rajshahi
BTCL	Bangladesh Telecommunications Company Limited	RU	University of Rajshahi
BU	University of Barishal	RUET	Rajshahi University of Engineering & Technology
BUBT	Bangladesh University of Business and Technology	SAU	Sylhet Agricultural University
BUET	Bangladesh University of Engineering and Technology	SBAU	Sher-e-Bangla Agricultural University
BUP	Bangladesh University of Professionals	SBMC	Sher-e-Bangla Medical College, Barishal
BUTEX	Bangladesh University of Textiles	SOMC	Sylhet MAG Osmani Medical College, Sylhet
CMC	Chittagong Medical College, Chittagong	SSHMC	Shaheed Suhrawardy Medical College, Dhaka
CoU	Cumilla University	SSKMC	Sheikh Sayera Khatun Medical College, Gopalganj
CU	University of Chittagong	SSMC	Sir Salimullah Medical College, Dhaka
CUET	Chittagong University of Engineering and Technology	SUST	Shahjalal University of Science and Technology
CVASU	Chattogram Veterinary and Animal Sciences University	TB	Terabyte
DDC	Dhaka Dental College, Dhaka	UGC	University Grants Commission of Bangladesh
DIU	Daffodil International University	VU	Varendra University
DMC	Dhaka Medical College, Dhaka		
DU	University of Dhaka		
DUET	Dhaka University of Engineering and Technology		
EU	Eastern University		
EWU	East West University		
HSTU	Hajee Mohammad Danesh Science & Technology University		
ICDDR'b	International Centre for Diarrhoeal Disease Research, Bangladesh		
ICMAB	The Institute of Cost and Management Accountants of Bangladesh		
ICSB	The Institute of Chartered Secretaries of Bangladesh		
IU	Islamic University, Bangladesh		
IUB	Independent University, Bangladesh		
IUBAT	International University of Business Agriculture and Technology		
IUT	Islamic University of Technology		
JKKNIU	Jatiya Kabi Kazi Nazrul Islam University		
JnU	Jagannath University		
JU	Jahangirnagar University		

